

Fair Career-Transition Pathways into the Green Economy for Displaced Fossil-Fuel Workers in EU Just Transition Regions

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Abstract

Career recommendations for displaced fossil-fuel workers in EU Just Transition regions sit at the intersection of three Sustainable Development Goals: decent work (SDG 8), gender equality (SDG 5), and climate action (SDG 13). We frame a content-based pathway recommender that, given a worker's current occupation in a carbon-intensive sector, returns ranked multi-step transitions toward green-economy occupations, with explanations grounded in skill overlap and a post-hoc fairness re-ranking on gender composition at the ISCO 2-digit level. Using the European Skills, Competences, Qualifications and Occupations classification (ESCO; ~3,000 occupations, ~13,900 skills) and Eurostat Labour Force Survey data, we observe that only 11% of ESCO occupations qualify as substantially green under the threshold $\text{greenShare} \geq 0.10$, that more than half of all occupations carry zero green skills, and that the ISCO classes containing the green occupations average 38.2% female employment against 43.6% in the rest—a 5.4 percentage-point structural skew. We then reflect on stakeholder tensions, the limits of demographic-parity re-ranking when the target landscape is itself segregated, and the impossibility of honest A/B evaluation when the user is a displaced worker. The report contributes a use case, a method concept, and a multi-stakeholder analysis grounded in EU-native data.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Machine learning*; • **Social and professional topics** → *Gender*.

Keywords

career recommendation, fairness, ESCO, just transition, sustainable development goals, multi-stakeholder recommender systems

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1 Introduction

The European Union's commitment to climate neutrality by 2050 will reshape its labour market. The Just Transition Fund, a €17.5 billion instrument under Cohesion Policy 2021–2027, supports approximately 108 NUTS-3 regions whose economies depend on coal, lignite, peat, oil shale, or carbon-intensive industries [3]. Workers in these regions face displacement on policy timetables: Polish lignite mines scheduled for closure by the early 2030s, Spanish coal-fired plants already shuttered, German hard-coal mining concluded in 2018. The political question of whether such transitions can be “just” depends in part on whether the workers in question can plausibly reach new employment in sectors aligned with climate goals [5].

Existing job and skill recommender systems are not well-suited to this question. They typically optimise for immediate match between a candidate's current profile and an available vacancy, not for multi-step skill-bridging pathways from a carbon-intensive occupation toward a green destination [2, 8]. The result is a system that may correctly inform a petroleum engineer that no immediate green role fits her profile, while saying nothing about which intermediate roles would, after one or two years of targeted retraining, place a wind-energy project manager position within reach.

In this report we frame, motivate, and reflect on a recommender system designed for exactly that gap. We do not build the system. We define the use case (Section 2), describe the data and method we would use, and then analyse stakeholder tensions, fairness trade-offs, evaluation realism, and broader societal challenges (Sections 3 and 4). The recommender we discuss is grounded in two SDGs in particular—decent work (SDG 8) and gender equality (SDG 5)—both implicated by climate action (SDG 13).

2 Use Case and Recommendation Task

2.1 Recommendation task

Given a worker currently employed in a fossil-fuel-intensive occupation in an EU NUTS-3 region designated under the Just Transition Fund, the system returns a ranked list of feasible multi-step career-transition pathways toward green-economy occupations. Each pathway is justified by skill overlap between consecutive steps; the path is constrained to be monotonically non-decreasing in greenness; and the final ranking is re-balanced to mitigate gender segregation at the destination ISCO class.

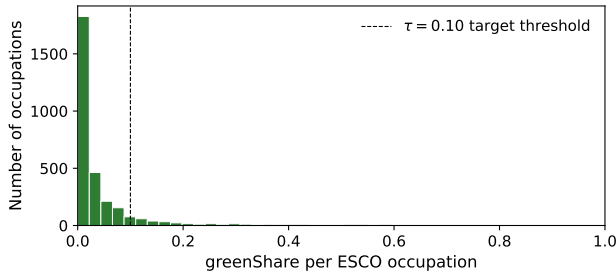


Figure 1: Distribution of greenShare across the 3,039 ESCO v1.2.1 occupations with a defined score. The dashed line marks the threshold $\tau = 0.10$ used to define the green-target set: 336 occupations (11.1%) cross it. More than half (53.6%) of all occupations carry zero green skills, and only 36 occupations (1.2%) cross the more restrictive $\tau = 0.50$.

Concretely: a 47-year-old maintenance technician at a Polish lignite-mining facility scheduled for closure by 2030 receives five ranked pathways. Each pathway shows the target occupation, intermediate occupations, the skills she already holds, the skills she would need to acquire, and which of those acquired skills count as “green” under ESCO’s classification. The displaceable population this addresses is concrete: Eurostat (lfsa_egan22d) records approximately 435,000 EU27 workers in 2025 in core fossil-fuel-extractive and refining sectors (NACE B05 coal and lignite mining, B06 crude petroleum and natural gas, B09 mining support, C19 manufacture of coke and refined petroleum), of which 178,000 are in coal and lignite alone.

2.2 SDG alignment

The recommendation task aligns with three Sustainable Development Goals. SDG 8 (decent work) is engaged by the question of whether displaced workers retain access to dignified employment as their original sector contracts. SDG 5 (gender equality) is engaged because, as we show in Section 3, the current green-occupation landscape is more male-dominated than the European labour market in aggregate; a recommender that maximises naive accuracy will accelerate this segregation. SDG 13 (climate action) is engaged structurally: the policy goal of climate neutrality is unattainable if the workforce cannot be moved into the sectors that deliver it.

2.3 Data foundation

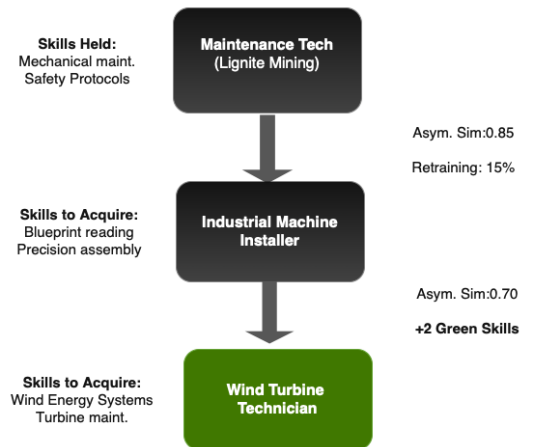
Our data foundation is the European Skills, Competences, Qualifications and Occupations classification, ESCO v1.2.1 [7]: approximately 3,000 occupations, 13,900 skills, an explicit essential/optional flag on each of approximately 126,000 occupation–skill links, an enumerated collection of 629 “green skills” added in later releases together with a derived per-occupation continuous greenShare score, and a mapping to ISCO 2- and 4-digit codes. Figure 1 shows the distribution of greenShare: 53.6% of ESCO occupations score zero, and only 11.1% (336 occupations) reach our chosen target threshold of 0.10. We use this looser threshold rather than the more restrictive 0.50 because the latter selects only 36 occupations dominated by environmental-policy and waste-management roles,

missing most of the renewable-energy, sustainable-construction, and electric-mobility occupations that the Just Transition narrative specifically targets. The threshold is itself a sensitivity parameter: $\tau = 0.05$ would admit 697 occupations and $\tau = 0.20$ only 152; admitting too liberally invites the greenwashing critique (a job is called “green” when only 5–10% of its skills are green) and admitting too conservatively excludes the trades the policy actually targets. A robustness analysis across thresholds is the academically honest move, which we flag as revision-phase work. Even at $\tau = 0.10$ the target landscape is structurally narrow.

Gender composition at the occupation level is not in ESCO. We complement it with Eurostat Labour Force Survey data lfsa_egai2d (employed persons by ISCO 2-digit class and sex), aggregated over EU27 for 2025 (Figure 2). A second LFS table, lfsa_egan22d, supplies the NACE 2-digit employment counts that scope the source-occupation pool (Section 2).

2.4 Method, in brief

Each occupation is represented as a sparse skill vector (essential skills weighted 1.0, optional 0.5). Pairwise asymmetric similarity defines the transition cost: for a source A and target B we use the fraction of B ’s skills that the worker already holds at A , which is asymmetric because $|A| \neq |B|$ in general. Other formulations are defensible—Dawson et al. [2] use a more elaborate weighted relative-mobility metric—but our choice is interpretable and aligns the similarity score directly with retraining effort: a score of 0.7 means 30% of the target’s essential skills must still be acquired. Pathways are generated by Dijkstra’s algorithm under a monotone-greenness constraint and a path length of at most four steps. As a worked illustration: a maintenance technician at a Polish lignite plant might receive a two-step path through “industrial machinery installer” to “wind turbine technician”, with the explanation listing the eight already-held skills and the four missing skills (of which two are tagged green by ESCO).



The top- K candidate paths are re-ranked with an objective that trades quality against demographic parity at the destination ISCO

2-digit class, with parameter λ swept to expose the trade-off. We did not build this system; this report describes the method conceptually and reflects on it. The course assignment grades that conceptual description and the reflection on it.

3 Reflection and Analysis

3.1 Stakeholders and the central tension

A career-transition recommender for displaced workers has many stakeholders: the worker herself; her trade union; the regional employment agency that will likely operate the touchpoint where she sees the recommendation; the retraining providers who will deliver the courses she is steered into; the green-sector employers who will eventually hire her; the regional and national governments answerable to electorates and EU climate-monitoring bodies; the European Commission as the funder of the Just Transition Fund; and—indirectly but importantly—future workers in regions that will receive labour migration if regional retraining underperforms.

The central tension, in our analysis, is between the worker’s individual autonomy and the systemic just-transition objective. A recommender that maximises individual fit—the highest-paying, fastest-reachable, lowest-disruption pathway—will, for any given source occupation, converge on a small number of destinations. Recall from Section 2 that only 336 ESCO occupations qualify as substantially green at $\tau = 0.10$, concentrated in a handful of professional and skilled-trades clusters; if the threshold is tightened to 0.50 the funnel shrinks to 36 occupations dominated by environmental-policy and waste-management roles. When millions of displaced workers receive recommendations from the same model, the labour market for the most-reachable green destinations becomes over-saturated; harder-to-reach but climate-critical destinations remain under-supplied; the regional employment agencies that operate the system are evaluated on placement metrics that reward steering workers into the easy targets; and the broader policy goal of distributed green workforce capacity is undermined.

This is a Mathew effect operating not at the user level (the conventional concern in recommender literature [9]) but at the labour-market level. Burke’s framework of multisided fairness [1] applies directly: at least three groups must be balanced—displaced workers as users, green-sector employers as item providers, regional governments as platform operators—each with different objective functions. Our method’s λ parameter is a single knob attempting to balance all of them, which is itself a design choice with consequences: a single global λ encodes a particular distributive theory, even if the operator never names it as such.

3.2 Method choices and what they assume

The method recap from Section 2 hides three load-bearing decisions. First, asymmetric similarity (shared skills over the target’s skill set) was chosen over symmetric cosine because the question “can a maintenance technician become a wind-turbine technician” is structurally different from its inverse. Anchoring the denominator on the target keeps the score interpretable as preparedness: a higher score means a smaller residual gap to acquire [2]. Second, we rejected three alternatives explicitly: collaborative filtering, because no real interaction data on cross-occupation transitions exists at EU scale; deep-learning embeddings, because 3,000 occupations is too

small to learn embeddings that outperform sparse skill features and because the resulting representations lose the per-recommendation explainability that justifies user trust; and large-language-model recommenders, because their opacity, lack of fairness guarantees, and lack of traceability to ESCO ground truth make them unsuited to a labour-market intervention with policy consequences.

Third, our explanation method is a literal set difference over the skill columns: “you already have these skills; you would need to acquire these, of which the following are tagged green by ESCO.” This is a strength relative to ranking models that produce uninterpretable scores, but it is also a vulnerability: the recommendation’s legibility depends entirely on ESCO’s classification quality. ESCO is a politically negotiated artefact; its “essential skills” for any occupation are subject to lobbying by training providers and professional bodies. We return to this attack surface in Section 3.5.

A fourth assumption hides in the pathway-generation constraint that greenness is monotonically non-decreasing along the recommended path. Real career transitions often pass through a “grey” bridge role—an industrial automation post, a generalist site-management role—that teaches a critical technical skill without itself being greener than the source. The monotone constraint excludes such bridge roles by construction, trading realism for narrative tractability. A more honest formulation would permit transient dips in greenness so long as the path-average or path-endpoint greenness exceeds the source; we keep the simpler constraint here and flag it as a limitation.

3.3 Fairness and the honest twist

The fairness argument depends on an empirical claim, which Figure 2 grounds. Across EU27 in 2025, the 32 ISCO 2-digit classes that contain at least one ESCO occupation with $\text{greenShare} \geq 0.10$ average 38.2% female employment, against 43.6% for the 10 ISCO classes without such an occupation (one class with zero recorded employment in 2025 is omitted)—a 5.4 percentage-point structural gap in the unweighted class average. Weighted by employment, the gap widens dramatically (42.3% versus 67.9%, since the non-green classes are dominated by care and clerical work with very large headcounts), but the unweighted measure is the more conservative claim about the structural distribution of green occupations across the labour market. The aggregate gap matters less than the class-level heterogeneity it conceals: several green-containing classes (skilled trades, electrical and electronic technicians, drivers and mobile plant operators) sit below 25% female, while a small number of green-containing service classes are at or above the EU labour-force average. The fairness problem is therefore not a uniform shift but a structural tilt, and it risks being amplified if a recommender steers female workers away from the few female-friendly green destinations and male workers into the male-dominated ones.

Naive demographic-parity re-ranking responds to this by penalising recommendations whose destination class deviates from 50% female. The honest twist is that this can degrade individual user satisfaction in ways that themselves matter morally. A male petroleum engineer who genuinely wants a wind-turbine engineering role can reasonably feel patronised by a system that nudges him toward a more “gender-balanced” destination class. A female worker with strong technical skills may equally resent being algorithmically

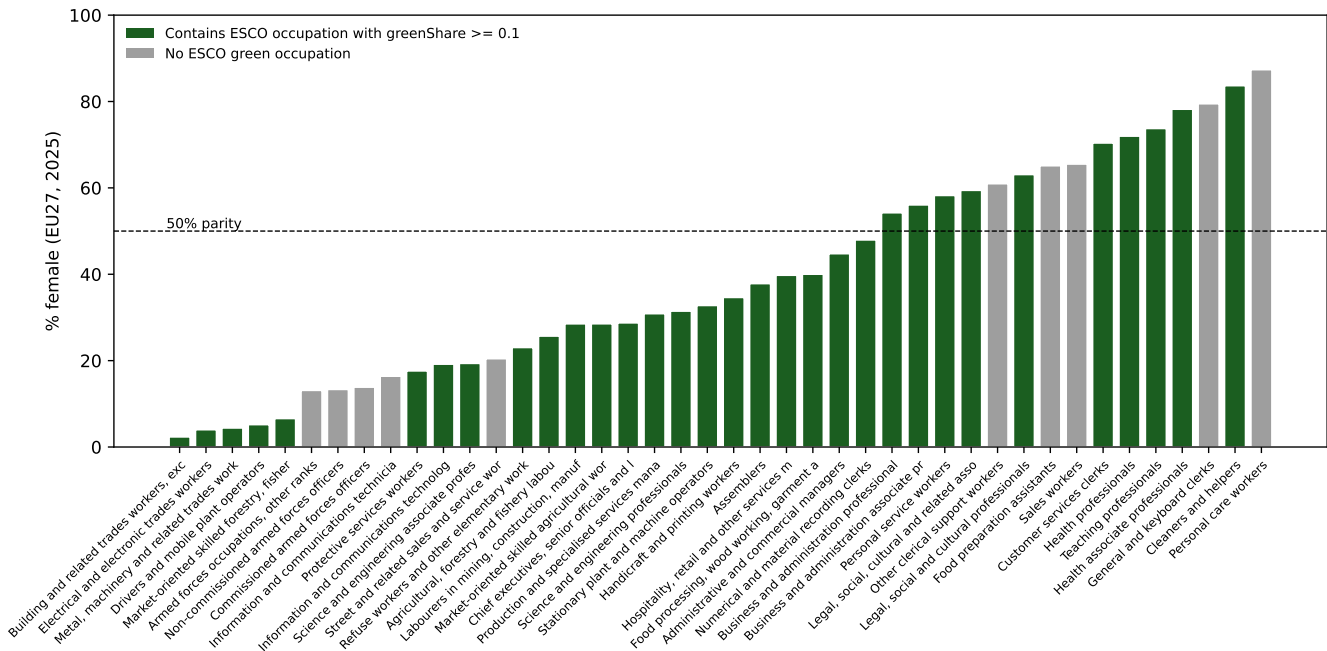


Figure 2: Share of female workers by ISCO 2-digit class, EU27 2025 (Eurostat 1fsa_egai2d; one class with zero recorded employment is omitted). Dark green: classes that contain at least one ESCO occupation with $\text{greenShare} \geq 0.10$. Across the 32 green-containing classes the unweighted average female share is 38.2%, against 43.6% for the 10 non-green classes—a 5.4 percentage-point structural gap. Class-level heterogeneity is wide: several green-containing classes (skilled trades, electrical and electronic technicians) sit below 25% female.

routed away from a male-dominated but well-paying green role for the sake of system-level parity statistics [6, 10].

System-level demographic parity, in other words, may be the wrong unit of analysis. The granularity mismatch between recommendation (made over ESCO’s ~3,000 occupations) and fairness measurement (taken over Eurostat’s 43 ISCO 2-digit classes) raises a Simpson’s-paradox risk: a green-containing ISCO class with apparent gender balance can conceal sub-class segregation in which women cluster in administrative or environmental-impact roles while men cluster in technical and engineering ones. Demographic parity satisfied at the 2-digit level can leave 4-digit segregation untouched or worsened. A more defensible objective might operate at the regional cohort level: ensure that, across the displaced workers in a single NUTS-3 region, the distribution of recommended destinations is no more gender-segregated than the existing distribution of underlying skills would predict. This is harder to formulate as a single λ , harder to compute, and harder to audit; but it shifts the moral burden from the individual recommendation to the population-level outcome, which is closer to the actual policy concern. A recommender that ships with a single global parity knob should be honest that it has solved a simpler, less appropriate problem [9].

3.4 Evaluation realism

The evaluation problem is the hardest part of this use case. We have no real interaction data: no logged clicks, no pathway completions, no outcomes. Standard offline ranking metrics;NDCG, recall, hit

rate—presuppose held-out interactions to evaluate against. Importing them here, as some pathway-recommendation papers do by treating proxy metrics as ground truth, is methodologically dishonest.

Three honest evaluation methods remain. First, skill-coverage as a feasibility proxy: for each recommended pathway, compute the fraction of the destination occupation’s essential skills already held by the source. This measures whether the pathway is reachable; it does not measure whether a real worker would take it. Second, Pareto-frontier analysis over λ : by sweeping the fairness re-ranking parameter and plotting average path quality against gender-parity deviation, we expose the trade-off explicitly rather than baking a particular operator preference into the deliverable. The Pareto curve is itself the evaluation artefact, not a single number [9]. Third, qualitative validation: present three worker personas (a Polish lignite-mining technician, a Spanish offshore-oil engineer, a German automotive assembly worker) and the resulting top-five recommended pathways to a panel of 10–20 peers, eliciting structured feedback on perceived feasibility, fairness, and trustworthiness on Likert scales.

The rubric for this assignment asks whether A/B testing or simulation could be used responsibly. For this use case, A/B testing on real displaced workers is mostly not appropriate. The cost of a poor recommendation is borne disproportionately by a vulnerable user (lost income during a closing window of state-funded retraining), the counterfactual is unobservable (we cannot rerun her career),

and the consent regime under which a regional employment agency could randomise its recommendations is unclear. A more responsible analogue is a slow regional roll-out with quasi-experimental comparison across NUTS-3 regions matched on labour-market fundamentals, with the recommender’s outputs made available alongside, not replacing, human caseworker guidance.

Distributional shift between training and production data is a documented failure mode [8]: ESCO is finer-grained than Eurostat ISCO 2-digit; the model is conceptually trained on EU-wide aggregates but would be deployed in a specific NUTS-3 region whose labour market may diverge sharply from those aggregates. A regional recommender deployed against EU-trained components is, in a real sense, always operating under partial covariate shift. A further evaluation gap is that skill coverage is only one dimension of feasibility; a deployable system would need to layer salary trajectory (Eurostat SES), regional vacancy density, commuting and relocation cost, and worker preference elicitation on top. The recommender as we describe it is a feasibility lens, not a complete decision-support tool, and adoption rates against a more holistic alternative would be the right outcome metric for any pilot.

3.5 Sustainability, explainability, cold start, manipulation

Four further dimensions deserve brief treatment. *Sustainability of the recommender itself* [4] is a modest concern: ESCO releases a major version roughly annually, and the sparse-skill-vector method imposes negligible training cost compared to deep models. Retraining cadence matches data cadence. *Explainability* is, as noted in Section 3.3, a strength of the chosen method: the per-recommendation rationale is a literal set difference of skills, with green-skill flags drawn directly from ESCO’s annotated collection. A real cost of the underlying representation is that ESCO encodes skill *presence* but not *proficiency*: a maintenance technician’s “mechanical maintenance” skill carries the same URI whether held at apprentice level or two decades into a senior role, so the similarity score over-credits superficial overlap and under-counts the depth a target occupation actually demands.

Cold start arises in two directions. New ESCO releases add emerging green occupations (the v1.2 release added roles in green hydrogen production and battery manufacturing), and these enter the target set with no historical recommendation footprint—they should be surfaced through diversity-aware exploration rather than waiting for organic uptake. In the opposite direction, occupations being phased out (deep coal mining, certain refining roles) should be archived rather than deleted: the workers currently in them remain in them, and a recommender that drops their source occupation entirely loses the ability to serve them at all.

Manipulation surface is the most interesting and least discussed risk. ESCO is a politically negotiated artefact maintained with input from member states and stakeholder bodies; the “essential skills” associated with any occupation are not natural facts but the outcome of governance processes susceptible to lobbying. A training provider whose certification is added to the essential-skills list for a high-demand green occupation gains a structural enrolment advantage that no fairness re-ranking will catch. ESCO governance, in short, becomes part of the recommender’s attack surface in a

way that is invisible to the recommender’s own metrics and to the worker who consumes its outputs. This argues for transparency: the version of ESCO in use, the criteria for inclusion of skills as essential, and the lineage of any updates should be auditable by the workers, unions, and journalists with the standing to scrutinise them [6].

4 Main Societal Challenges

If the recommender works at scale—say, 100,000 EU workers redirected over five years—three second-order effects become salient. *Regional skill clustering*: workers within a single Just Transition region, served by the same recommender and the same retraining-provider catalogue, converge on similar destinations, recreating the sectoral concentration that made the original displacement painful in the first place. *Retraining-provider lock-in*: the small set of providers offering certifications for the small set of green destinations gain a quasi-monopolistic position. *ISCO oversupply imbalance*: certain ISCO 2-digit classes attract many more new entrants than the labour market can absorb, depressing wages in the very destinations the recommender promotes as economically attractive.

Who operates the recommender materially shapes which of these effects dominate. A state agency operates under public-procurement rules and democratic accountability for placement statistics; its λ tilts toward whatever the agency is being measured on, which historically has been short-term placement count. A private platform monetises engagement and, indirectly, the retraining providers it surfaces; its λ tilts toward retraining-revenue maximisation. A union or works-council consortium operates with explicit attention to wage compression and collective bargaining leverage; its λ tilts away from individual fast-placement metrics and toward sectoral labour-market structure. The same algorithmic skeleton produces meaningfully different societal outcomes depending on which party owns it.

Geography compounds these dynamics. ESCO is multilingual, but the EU labour market is not: a Polish lignite-mining technician has limited labour-market access in Spain or Portugal regardless of skill alignment. A pathway that is technically optimal across the EU may be effectively unreachable within her actual mobility envelope. This argues for explicit regional masking, but masking has its own fairness implications: if the within-region green destinations are scarce or low-paid, regional masking locks workers into worse outcomes.

A recommender deployed under the banner of EU climate policy will likely be claimed as a tool of that policy. But the system is not validated against actual emissions outcomes, retention in green roles, or career trajectory beyond first placement. Without that validation [4], it is, strictly, a job recommender with green branding. The honest deployment posture is to say so.

5 Conclusion

We framed a content-based pathway recommender for displaced fossil-fuel workers in EU Just Transition regions as a use case for reflection. The central insight is that fairness in this setting cannot be reduced to a global demographic-parity knob: when the target landscape is itself narrow (only 336 of 3,039 ESCO occupations are

substantially green) and structurally segregated (a 5.4 percentage-point lower female share in green-containing ISCO classes), naive system-level re-ranking trades against individual user satisfaction in ways that warrant explicit acknowledgement, and the unit of analysis arguably belongs at the regional cohort level rather than the individual recommendation. The absence of real interaction data forces honest evaluation through skill-coverage proxies, Pareto-frontier exposure of the fairness trade-off, and qualitative panels rather than dishonestly imported offline ranking metrics. The reflection process itself surfaced tensions—labour-market Mathew effects, ESCO’s governance attack surface, the operator-dependent meaning of λ —that a pure-accuracy framing would have hidden. Future work would extend to intersectional analyses, regional NUTS-2 LFS overlays, and a controlled pilot with a national employment agency.

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A Generative AI Disclosure

We used generative AI tools as supportive instruments during the work in this report. All decisions regarding the different parts like problem framing, recommendation tasks, method choice, fairness discussion etc. were made by us the authors. Every suggestion made by AI tools was reviewed, edited appropriately and checked by the other group members. Authors of this report are responsible for the content of it.

B Organization of Group Work

Our group collaborated through the topic selection process, where we reviewed the possible topics and chose the one closest to our interests. The work began by discussing the SDG-linked recommended domains and later selected the topic. We selected a topic regarding job market specifically named: "Fair Career-Transition Pathways into the Green Economy for Displaced Fossil-Fuel Workers in EU Just Transition Regions" as we felt it offered a strong connection among recommender systems, fairness, and societal impact.

Although responsibilities were divided between us, all major decisions were made collectively. Draft sections were reviewed by the full group and we discussed disagreements about method assumptions, fairness objectives, and evaluation realism.